

Unit 4: Energy Science & Introduction to Nanomaterials

Energy Science

Fuels

- **Fuels** –Introduction, definition, classification, calorific value, Gross and Net, theoretical calculation.
- **Solid fuel-** Coal- classification, Analysis- Proximate and ultimate analysis

INTRODUCTION

- A fuel is a combustible substance containing carbon as the main constituent which on proper burning gives large amount of heat that can be used economically for domestic and industrial purposes.
- During the process of combustion of a fuel, the atoms of carbon, hydrogen, etc combine with oxygen with simultaneous liberation of heat. The calorific value of a fuel depends mainly on the two elements.

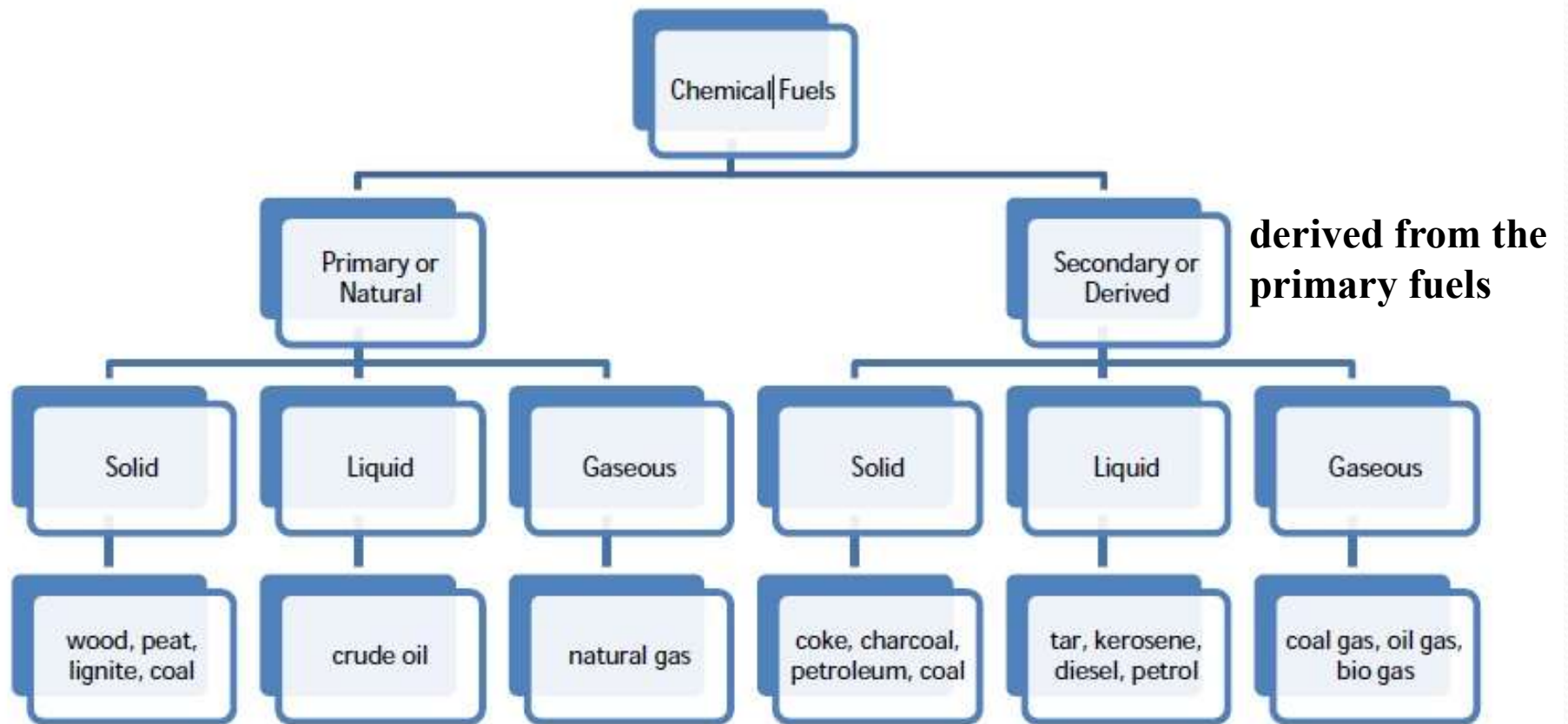


- So, carbon compounds have been used for many centuries as the source of heat and energy.
- The main source of fuel is coal and petroleum.
- These are stored fuels available in earth's crust and are generally called fossil fuels because they were formed from the fossilised remains of plants and animals.

Characteristics of Good Fuels:

- High Calorific Values
- Moderate Ignition Temperature
- Low Moisture Content
- Low Ash Content
- Moderate Velocity of Combustion
- Should not produce harmful products
- Low Cost
- Easy Storage & Transportation
- Easily Controllable

Classification of Fules



CALORIFIC VALUE

- Calorific value of a fuel is "the total quantity of heat liberated, when a unit mass (or volume) of the fuel is burnt completely."

Units of heat :

- (1) '**Calorie**' is the amount of heat required to raise the temperature of **one gram of water through one degree Centigrade**.
- (2) "**Kilocalorie**" is equal to 1,000 calories. It may be defined as 'the quantity of heat required to raise the temperature of **one kilogram of water through one degree Centigrade**.
Thus: $1 \text{ kcal} = 1,000 \text{ cal}$
- (3) "**British Thermal unit**" (B.T.U.) is defined as "the quantity of heat required to raise the temperature of **one pound of water through one degree Fahrenheit** (60-61°F). This is the English system unit.
 $1 \text{ B.T.U.} = 252 \text{ cal} = 0.252 \text{ kcal}$ $1 \text{ kcal} = 3.968 \text{ B.T.U.}$

HCV and LCV

Higher or gross calorific value:

- Gross or higher calorific value (HCV) is "the total amount of heat produced, when unit mass/volume of the fuel has been burnt completely and the **products of combustion have been cooled to room temperature**"(i.e., 15°C or 60°F).

Lower or net calorific value (LCV)

- Net or lower calorific value (LCV) is "the net heat produced, when unit mass /volume of the fuel is burnt completely and the **products are permitted to escape**".
- Net calorific value= Gross calorific value - Latent heat of condensation of water vapour produced
- $$= \text{GCV} - \text{Mass of hydrogen per unit weight of the fuel burnt} \times 9 \times \text{Latent heat of condensation of water vapour}$$

- **Dulong's formula** for calorific value from the chemical composition of fuel is :
- **HCV** = $\frac{1}{100} [8,080 C + 34,500 (H - O/8) + 2,240 S]$ kcal/kg
- where C, H, O, and S are the percentages of carbon, hydrogen, oxygen and sulphur in the fuel respectively.
- In this formula, oxygen is assumed to be present in combination with hydrogen as water, and

Relation between LCV & HCV:

- **LCV** = $[HCV - 9H/100 \times 587]$ kcal/kg = $[HCV - 0.09 H \times 587]$ kcal/kg
- This is based on the fact that 1 part of H by mass gives 9 parts of H₂O, and latent heat of steam is 587 kcal/kg.

calculate the gross & net calorific value of coal having following compositions:

Carbon = 85% Hydrogen = 8% Sulphur = 1%, Nitrogen = 2%
Ash = 4%

Latent heat of combustion of steam = 587 Kcal/kg

According to Dulong's formula -

$$\text{HCV} = \frac{1}{100} \left[8080C + 34500 \left(H - \frac{O}{8} \right) + 2240S \right] \text{Kcal/kg}$$

Given = C = 85%, H = 8%, S = 1%.

Nitrogen of ash, don't contribute to the calorific value.

$$HCV = \frac{1}{100} \left[8080 \times 85 + 34500 \left(8 - \frac{0}{8} \right) + 2240 \times 1 \right]$$

$$= \frac{1}{100} [686800 + 276000 + 2240] \text{ ~~cal/g~~ kcal/kg}$$

$$= \frac{1}{100} [965040] = \underline{9650.4 \text{ ~~cal/g~~ kcal/kg}}$$

$$\text{net caloric value (NCV)} = (HCV - 0.09H \times 587) \text{ kcal/kg}$$

$$= 9650.4 - 0.09 \times 8 \times 587 \text{ kcal/kg}$$

$$= \underline{9227.8 \text{ kcal/kg}}$$

② A coal has the following composition by weight:

C = 90%, O = 8%, S = 0.5%, N = 0.5% & Ash = 2.5%

Net calorific value of the coal was found to be 8,490.5 kcal/kg.

Calculate the percentage of hydrogen of higher calorific value of coal.

$$\begin{aligned}\rightarrow \text{HCV} &= (\text{HCV} + 0.09\text{H} \times 587) \text{ kcal/kg} \\ &= (8490.5 + 0.09\text{H} \times 587) \text{ kcal/kg} \\ &= (8490.5 + 52.8\text{H}) \text{ kcal/kg.} \quad \text{--- ①}\end{aligned}$$

$$\text{Also, } \text{HCV} = \frac{1}{100} \left[8080 \times 90 + 34500 \left[\text{H} - \frac{3 \cdot 0}{8} \right] + 2240 \times 0.5 \right] \text{ kcal/kg}$$

$$= [7272 + 345\text{H} - 129.4 + 11.2] \text{ kcal/kg}$$

$$= [7754.8 + 345\text{H}] \text{ kcal/kg} \quad \text{--- ②}$$

From ① & ② we get -

$$7754.8 + 345H = 8490.5 + 52.8H$$

or $292.2H = 8490.5 - 7754.8 = 1335.7$

or % of H = ~~1335.7~~ $1335.7 / 292.2 = \underline{4.575\%}$

$$\therefore \text{HCV} = (8490.5 + 52.8 \times 4.575) \text{ kcal/kg}$$

$$= 8490.5 + 241.3 \text{ kcal/kg}$$

$$= \underline{\underline{8731.8 \text{ kcal/kg}}}$$

Analysis of Coal Sample

- In order to assess the quality of coal the following two types of analysis are made.
- **1. Proximate Analysis** It includes the determination of **moisture, volatile matter, ash and fixed carbon**.
- This gives quick and valuable information regarding commercial classification and determination of suitability for a particular industrial use.



2. Ultimate Analysis It includes the determination of **carbon, hydrogen, nitrogen, sulphur and oxygen** in coal. Since it is used

For the determination of elements present in the coal, it is also called **elemental analysis**.

This analysis gives exact results and are useful in calculating the calorific value of coal using Dulong's formula.

Proximate analysis

- (1) **Moisture**: About 1 g of finely powered air-dried coal sample is weighed in a crucible. The crucible is placed inside an electric hot air-oven, maintained at 105° - 110°C. The crucible is allowed to remain in oven for 1 hour and then taken out, cooled in a desiccator and weighed. Loss in weight is reported as moisture (on percentage-basis).

- $$\text{Percentage of moisture} = \frac{\text{Loss in weight}}{\text{Wt. of coal taken}} \times 100$$

Moisture lowers the effective calorific value of coal.

- (2) **Volatile matter**: The dried sample of coal left in the crucible in (1) is then covered with a lid and placed in an electric furnace (muffle furnace), maintained at 925°± 20°C. The crucible is taken out of the oven after 7 *minutes* of heating. The crucible is cooled first in air, then inside a desiccator and weighed again. Loss in weight is reported as volatile matter on percentage-basis.

Percentage of volatile matter=

$$\frac{\text{Loss in weight due to removal of volatile matter}}{\text{Wt. of coal sample taken}} \times 100$$

A high volatile matter containing coal burns with a long flame, high smoke and has low calorific value.

- (3) **Ash:** The residual coal in the crucible in (2) is then heated without lid in a muffle furnace at $700 \pm 50^{\circ} \text{C}$ for $1/2$ hour. The crucible is then taken out, cooled first in air, then in desiccator and weighed.

- Heating, cooling and weighing is repeated, till a *constant weight* is obtained. The residue is reported as ash on percentage-basis.

- $$\text{Percentage of ash} = \frac{\text{Wt. of ash left}}{\text{Wt. of coal taken}} \times 100$$
-

- **Ash is a useless, non-combustible matter, which reduces the calorific value of coal.**

(4) **Fixed carbon:** Percentage of fixed carbon = $100 - \% \text{ of (moisture + volatile matter + ash)}$

- high percentage of fixed carbon is desirable.
 - **Higher the percentage of fixed carbon, greater is its calorific and better the quality coal.**

Ultimate analysis

Carbon and hydrogen: About 1-2 g of accurately weighed coal sample is burnt in a current of oxygen in a combustion apparatus. C and H of the coal are converted into CO₂ and H₂O respectively. The gaseous products of combustion are absorbed respectively in KOH and CaCl₂ tubes of known weights. The increase in weights of these are then determined.

- $$\text{Percentage of C} = \frac{\text{Increase in weight of KOH tube} \times 12 \times 100}{\text{Weight of coal sample taken} \times 44}$$

- $$\text{Percentage of H} = \frac{\text{Increase in weight of CaCl}_2 \text{ tube} \times 2 \times 100}{\text{Weight of coal sample taken} \times 18}$$

- Greater the percentage of carbon and hydrogen better is the coal in quality and calorific value.

Nitrogen: About 1 g of accurately weighed powdered coal is heated with concentrated H_2SO_4 along-with $\text{K}_2\text{S}_2\text{O}_8$ (catalyst) in a long-necked flask (called *Kjeldahl's flask*). After the solution becomes clear, it is treated with excess of KOH and the liberated ammonia is distilled over and absorbed in a *known volume of standard acid solution*. The unused acid is then determined by back titration with standard NaOH solution. From the volume of acid used by ammonia liberated, the percentage of N in coal is calculated as follows:

- Percentage of N =
$$\frac{\text{Volume of acid used} \times \text{Normality} \times 1.4}{\text{Weight of coal taken}}$$
- Nitrogen has no calorific value and hence, its presence in coal is undesirable; thus, a good quality coal should have very little nitrogen content.

Sulphur is determined from the washings obtained from the known mass of coal, used in a bomb calorimeter for determination of a calorific value. During this determination, S is converted into sulphate. The washings are treated with barium chloride solution, when barium sulphate is precipitated. This precipitate is filtered, washed and heated to constant weight.

- Percentage of S =
$$\frac{\text{Weight of BaSO}_4 \text{ obtained} \times 32 \times 100}{\text{Weight of coal sample taken in bomb} \times 233}$$

Ash determination is carried out as in proximate analysis.

Oxygen: It is obtained by difference.

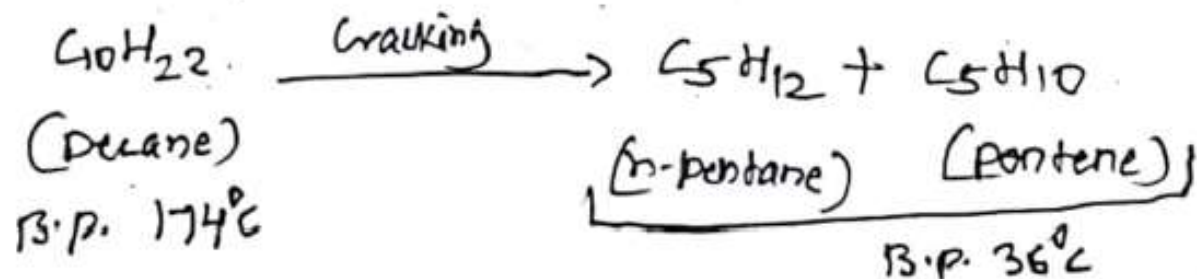
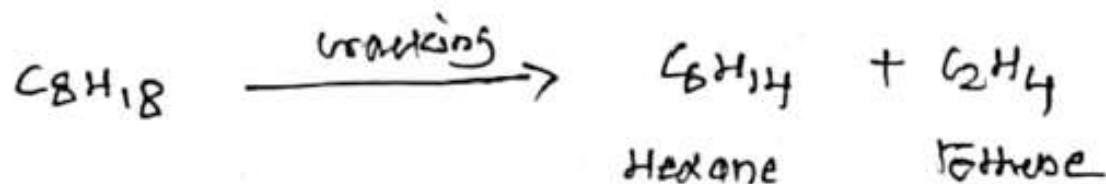
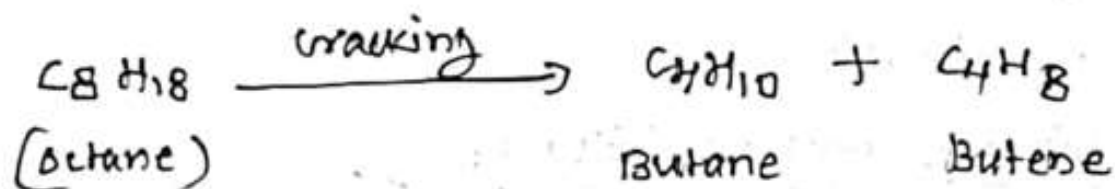
- Percentage of O = $100 - \text{Percentage of (C + H + S + N + ash)}$
- Oxygen content decreases the calorific value of coal.

Cracking of Oil

- The decomposition of bigger hydrocarbon molecules into simpler, low boiling hydrocarbons of lower molecular weight is called cracking.
- The gasoline obtained from the fractional distillation of petroleum, has the highest demand as a motor fuel, but the yield of this fractions is only 20-30% (Crude oil) and also quality as straight-run gasoline which is not good an_ hence is used only after proper blending.
- To overcome these difficulties, the higher boiling fractions (e.g. fuel oil and gas oil) are converted into lower boiling fractions gasoline (petrol) by cracking process.
- The cracked gasoline gives better engine performance i.e., they are suitable for spark -ignition engines of automobiles.
- In cracking process, higher saturated hydrocarbon molecules are converted into simpler molecules such as paraffinic and olefinic hydrocarbons,
- There are two methods of cracking in use
- **1. Thermal cracking 2. Catalytic cracking**

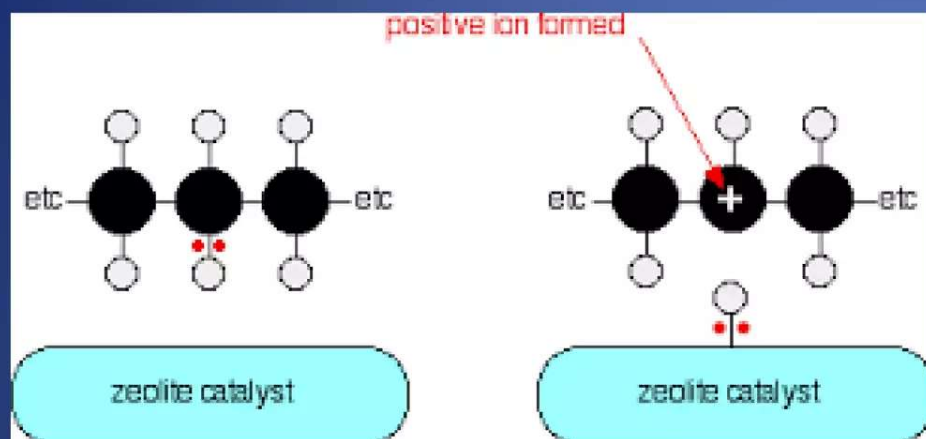
cracking of oils:

→ When long chain alkanes are heated well above their boiling points ($400-800^{\circ}\text{C}$) with or without catalyst, they are broken into smaller alkenes & H_2 . Such thermal decomposition with reference to petroleum hydrocarbon is called cracking.



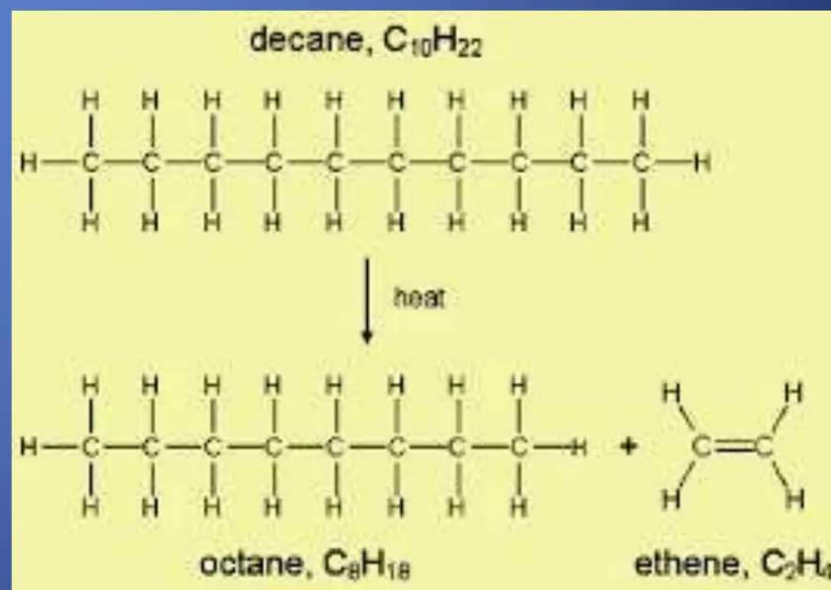
There are 2 methods of cracking:

➤ TYPE OF CRACKING:-

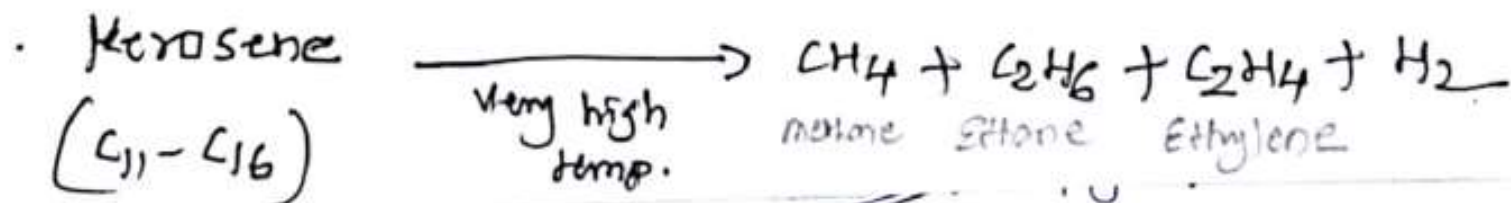


1.) Catalytic Cracking

2.) Thermal Cracking

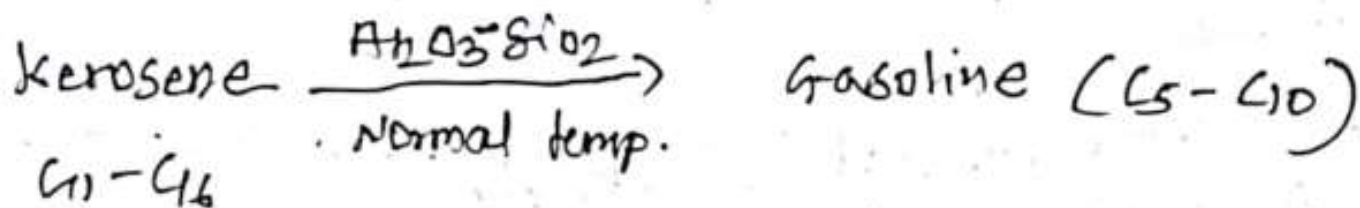


① Thermal cracking - It is carried out at high temp. & pressure.



② Catalytic cracking - In presence of suitable catalyst, cracking can be done at lower temp.

→ The catalyst are Alumina (Al_2O_3), Silica (SiO_2), etc.



Knocking

- Knocking is a kind of **explosion** due to rapid pressure rise occurring in an IC engine.
- In a petrol engine, a mixture of gasoline vapour and air at 1: 17 ratio is used as fuel. This mixture is compressed and ignited by an electric spark.
- The products of oxidation reaction (combustion) increases the pressure and pushes the piston down the cylinder.
- If the combustion proceeds in a regular way, there is no problem in knocking.
- But in some cases, the rate of combustion (oxidation) will not be uniform due to unwanted chemical constituents of gasoline.
- The rate of ignition of the fuel gradually increases and the final portion of the fuel-air mixture gets ignited instantaneously producing an explosive sound known as "Knocking".
- Knocking property of the fuel reduces the efficiency of engine. So a good gasoline should resist knocking.

Antiknock agents:

those compounds which when added to gasoline or petrol increases its octane number & decreases knocking are called antiknock ~~agent~~ agent / compounds. They improve the quality of petrol.

Examples: — i) Tetraethyl lead $[(C_2H_5)_4Pb]$ (TEL) → TEL - causes serious pollution
ii) Diethyl telluride $[(C_2H_5)_2Te]$ ↓,
iii) Iron carbonyl $[Fe(CO)_5]$ Methyl Tertiary-Butyl ether (MTBE).

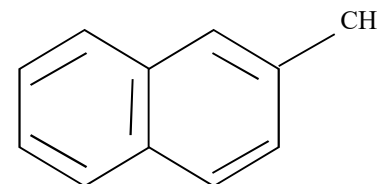
OCTANE NUMBER (or) OCTANE RATING

- Octane number is introduced to express the knocking characteristics of petrol.
 - It has been found that **n-heptane knocks very badly** and hence, its anti-knock value has been given **zero**.
 - On the other hand, **iso-octane gives very little knocking** and so, its anti-knock value has been given **100**.
 - *Thus octane number is defined as 'the percentage of iso-octane present in a mixture of iso-octane and n-heptane.'*
-
- **Iso-octane (Octane number = 100)**
 - $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$
 - **n - heptane (Octane number = 0)**

CETANE NUMBER (or) CETANE RATING

- **Cetane number is introduced to express the knocking characteristics of diesel.**
- Cetane has a very short ignition lag and hence its cetane number is taken as 100.
- On the other hand 2-methyl naphthalene has a long ignition lag and hence its cetane number is taken as zero.

- $\text{CH}_3 - (\text{CH}_2)_{14} - \text{CH}_3$
- n-cetane (hexa decane)
- cetane number = 100



- **2-methyl naphthalene (cetane number = 0)**
- Thus the cetane number is defined as "**the percentage of hexa decane present in a mixture of hexa decane and 2-methyl naphthalene, which has the same ignition lag as the fuel under test**".
- The cetane number decreases in the following order.
- n-alkanes > Cycloalkanes > alkenes > branched alkanes > aromatics
- The cetane number of a diesel oil can be increased by adding additives called dopes.